MedRAG: Medical Research Assistant

Project Report and Technical Documentation

1. Introduction

Medical knowledge is frequently stored in lengthy PDF documents such as clinical guidelines, drug monographs, diagnostic summaries, and research publications. Although these PDFs contain essential information, locating specific details within them is often slow and inefficient. Traditional keyword search tools only return word matches without understanding context, and manually scanning large documents makes it easy to miss important information.

The MedRAG (Medical Research Assistant) system was developed to address this challenge. Instead of relying on a language model's general training data, MedRAG retrieves and uses excerpts from the user's own uploaded medical PDFs as the foundation for its answers. The system integrates Retrieval-Augmented Generation (RAG) with a multi-agent refinement pipeline, ensuring that responses are accurate, contextual, and grounded in the provided documents.

MedRAG is built around several coordinated components: a Streamlit-based interface, a FastAPI backend, a PDF processing pipeline for extracting and chunking text, Sentence-BERT embeddings for semantic representation, a FAISS vector index for efficient retrieval, and specialized agents for classification, answer generation, and validation. The goal is to deliver trustworthy, document-backed answers to medical research questions.

2. Problem Definition

Medical PDFs present several obstacles that limit their usefulness for quick question answering:

- They are long, densely written, and formatted inconsistently.
- Complex layouts and technical language make them difficult to navigate.
- Important content may appear across multiple sections that cannot be easily skimmed.
- Search functions lack context and often fail to surface meaningful information.

Traditional question-answering systems also struggle with this format. Large language models cannot ingest entire documents due to context limitations, and they lack the ability to identify which parts of a PDF are relevant to a specific question. Without retrieval, an LLM may overlook critical details or generate responses based on its training data rather than the user's uploaded content.

Because medical information must be precise and verifiable, users need a system that can quickly locate relevant text in complex PDFs and present it in a clear, accurate way. This motivates the need for a structured, retrieval-based system that uses the document itself as the basis for its answers.

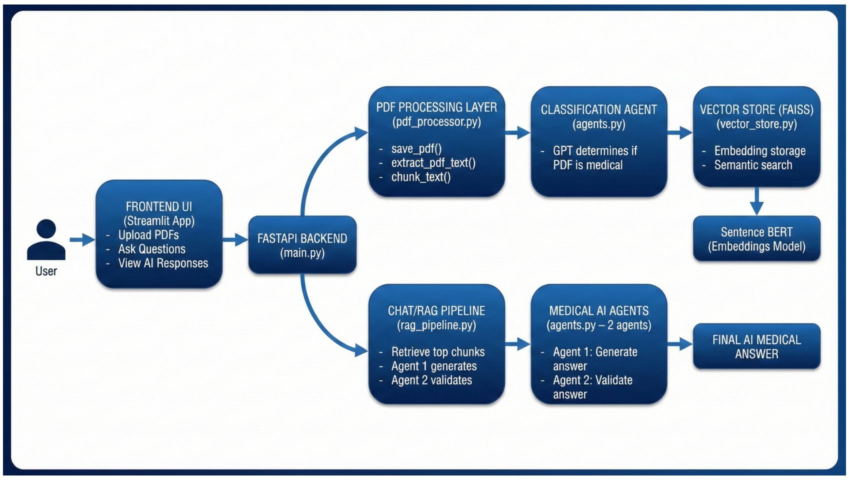
3. System Overview

MedRAG addresses these challenges through a four-stage workflow:

1. **Document Processing:** Extract and segment text into meaningful chunks.
2. **Semantic Retrieval:** Embed and index chunks using Sentence-BERT and FAISS.
3. **Answer Generation:** Produce an initial response from retrieved excerpts.
4. **Answer Validation:** Refine the response to ensure accuracy and clarity.

This pipeline ensures that every answer originates from the uploaded documents, not from the model's assumptions.

4. System Architecture



The MedRAG workflow follows a clear sequence of steps:

4.1. User Input

The user uploads medical PDFs and enters a question using the Streamlit interface.

4.2. Backend Coordination

FastAPI receives the inputs and passes them to the processing pipeline.

4.3. PDF Processing

The system extracts text and divides it into segments that can be embedded and retrieved efficiently.

4.4. Document Classification

A classification agent determines whether each document contains medical content and filters out irrelevant files.

4.5. Embedding and Indexing

Sentence-BERT converts chunks into semantic embeddings, which are stored in a FAISS index.

4.6. Retrieval

When a question is submitted, the system embeds the query and retrieves the most relevant chunks.

4.7. Answer Generation and Validation

A generator agent produces an initial answer from the retrieved excerpts, and a validator agent reviews and refines it for clarity and correctness.

4.8. Response Delivery

The final answer is displayed to the user through the Streamlit interface.

This modular design ensures clear separation of responsibilities and improves maintainability and reliability.

5. Key Components

5.1. User Interface (Streamlit)

The user interface provides a simple and intuitive way to upload PDFs and submit questions. All computational work takes place on the backend, keeping the interface responsive and easy to use.

5.2. FastAPI Backend

FastAPI serves as the system's central controller. It manages file uploads, triggers PDF processing, generates embeddings, performs retrieval operations, and coordinates the multi-agent answering pipeline.

5.3. PDF Processing

The processing layer prepares each document for retrieval. It includes the following functions:

- `save_pdf()` – stores uploaded files

- `extract_pdf_text()` – extracts text page by page
- `chunk_text()` – divides extracted text into coherent, meaningful segments

5.4. Classification Agent

A classification agent evaluates whether the document contains medical content, filtering out irrelevant material and improving retrieval quality.

5.5. Embeddings and Vector Search

MedRAG uses:

- Sentence-BERT to generate semantic embeddings
- FAISS to store vectors and perform fast similarity search

5.6. RAG Pipeline

The retrieval workflow consists of three steps:

1. Embedding the user's question
2. Searching the FAISS index to identify the most relevant chunks
3. Passing these retrieved segments to the answering agents

6. Multi-Agent Answering System

MedRAG uses a structured multi-agent pipeline to generate accurate, well-supported answers.

6.1. Classification Agent

Determines whether uploaded documents contain medical content.

6.2. Generator Agent

Produces an initial draft response using retrieved text chunks.

6.3. Validator Agent

Checks consistency with the retrieved text and refines the response for clarity and completeness.

7. Design Decisions

Key design decisions include:

- Retrieval-Augmented Generation (RAG) ensures answers stay grounded in uploaded PDFs.
- Sentence-BERT embeddings offer computational efficiency with strong semantic similarity.
- FAISS indexing supports fast and scalable similarity search.
- Streamlit and FastAPI separate the interface from backend logic.
- Multi-agent reasoning improves answer reliability through a generate–validate pipeline.

8. Results

Across multiple tests using drug leaflets, clinical excerpts, and guideline documents, MedRAG demonstrated:

- Accurate retrieval of relevant document sections
- Clear and well-grounded generated answers
- Reduced hallucinations due to validation step
- Smooth user workflow for uploading documents and asking questions

9. Limitations

- Scanned PDFs require OCR to be processed.
- Tables and figures do not extract cleanly.
- Page-level citations are not yet included.
- Only text content is processed; images are ignored.

10. Conclusion

MedRAG demonstrates how retrieval, semantic search, and multi-agent reasoning can be combined to deliver accurate, document-grounded responses for medical research tasks. Its modular architecture provides a strong foundation for future enhancements and practical deployment.